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Broadband negative refraction based on a pair of anti-parity-time structures

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Abstract – An anti-parity-time (A-PT) symmetric system, of which the refractive index satisfies $n(-x) = -n^*(x)$, can present intriguing scattering properties like continuous coherent perfect absorption (CPA) spectrum and laser spectrum, which are symmetric to each other in the parameter space. In this work, an acoustic parity-time (PT) symmetric system consisting of two A-PT symmetric structures is introduced to realize a broadband negative refraction. When the two A-PT symmetric structures are at their CPA mode and laser mode, respectively, the PT symmetric system can display its unidirectional negative refraction property. As the two A-PT structures we design here have a broadband CPA and laser modes, the PT symmetric negative refraction system will be broadband. Besides, the exceptional point (EP) of the proposed PT symmetric system corresponds to the CPA point and laser point of its two components, and it is unidirectionally transparent on broadband as well.

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Introduction. – Manipulating acoustic waves with artificial materials (“metamaterials”) is always a challenging project for the researchers. In the last few years, there have been great advances in the study on acoustic metamaterials, like double negative materials [1–3], acoustic wave clocking [4,5], acoustic capsule [6], coding metasurface [7–12], acoustic metalens [13–15], metamaterial absorbers [16–21] and so on. The concept of parity-time (PT) symmetry, which is originated from the non-Hermitian quantum theory [22], has been introduced to optics, electromagnetics and acoustics because of the analogy between the classic wave equation with the Schrödinger equation [23–33]. The acoustic PT symmetric system can be realized based on the materials with balanced gain and loss [23–28], or only loss to get an easier access to experimental realization [29–31], and has showcased a lot of different intriguing properties, such as unidirectional invisibility, acoustic logic gate, coherent perfect absorption (CPA) and laser ef-

fect, asymmetric diffraction, single-sided beam splitting and negative refraction. Recently, a new class of anti-parity-time (A-PT) symmetric structures [34–40], which displays some interesting properties, such as bidirectional total transmission, continuous laser and CPA spectrum, is proposed based on the negative-index material (NIM) and positive-index material (PIM).

In this paper, we propose a broadband negative refraction system based on two A-PT symmetric structures at their CPA (laser) modes by applying balanced PIM and NIM with gain and loss. With further study, we find that the density and refractive index distributions of the total negative refraction system satisfy the conditions of PT symmetry, *i.e.*, $\rho(-x) = \rho(x)$ and $n(-x) = n^*(x)$, and the exceptional point of the total PT symmetric system corresponds to the CPA point and the laser point of the two separate A-PT symmetric structures. Different from the negative refraction system based on the CPA-laser point or the exceptional point (EP) of the PT symmetric material [41–44], the system we design can be broadband because the CPA and laser modes of A-PT

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symmetric structures are continuous in the parameter space.

Calculation of the scattering properties. – As shown in fig. 1(a), the 1D A-PT symmetric structure we design is composed of a pair of positive-index material and negative-index material, each of them with thickness $l = 50$ mm, placed in the passive background medium air, with a density of $\rho_0 = 1.21$ kg/m³ and a sound velocity of $c_0 = 343$ m/s. To derive the sound pressure distribution (p) in the background medium, the transfer matrix method (TMM) is used (see details in the Supplementary Material [SupplementaryMaterial.pdf](#) (SM)). As a plane harmonic wave propagates through the structure, p can be expressed as

$$p(x) = \begin{cases} p_{+L}e^{ik_0(x+l)} + p_{-L}e^{-ik_0(x+l)}, & x < -l, \\ p_{+R}e^{ik_0(x-l)} + p_{-R}e^{-ik_0(x-l)}, & x > l, \end{cases} \quad (1)$$

where the subscript $\pm$ denotes the acoustic wave traveling forward and backward, respectively, in the left (L) and right (R) regions. $k_0 = \frac{\omega}{c_0}$ is the wave number in the background medium air and $\omega = 2\pi f$ (f is the frequency) is the circular frequency. The time factor $e^{-j\omega t}$ has been dropped. The acoustic pressure in the left and right regions can be connected by a transfer matrix $T = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}$,

$$\begin{pmatrix} p_{+R} \\ p_{-R} \end{pmatrix} = T \begin{pmatrix} p_{+L} \\ p_{-L} \end{pmatrix}. \quad (2)$$

Based on the continuity conditions of the acoustic pressure and normal velocity at the interfaces, the elements of the transfer matrix T can be derived. Originated from reciprocity of this system, the determinant of T is always 1, *i.e.*, $\det(T) = T_{11}T_{22} - T_{12}T_{21} = 1$, which is calculated in the SM. Accordingly, the transmission (t) and reflection (r_L and r_R) coefficients of the structure can be derived by imposing that $p_{+L} = 0$ or $p_{-R} = 0$. They can be expressed as

$$r_L = -\frac{T_{21}}{T_{22}}, \quad r_R = \frac{T_{12}}{T_{22}}, \quad t = \frac{T_{11}T_{22} - T_{12}T_{21}}{T_{22}} = \frac{1}{T_{22}}. \quad (3)$$

With t , r_L and r_R , the scattering matrix S of the structure can be expressed as

$$S = \begin{pmatrix} t & r_L \\ r_R & t \end{pmatrix}, \quad (4)$$

which can relate the acoustic waves of the two sides of the structure by $\begin{pmatrix} p_{-L} \\ p_{+R} \end{pmatrix} = \begin{pmatrix} t & r_L \\ r_R & t \end{pmatrix} \begin{pmatrix} p_{-R} \\ p_{+L} \end{pmatrix}$. The eigenvalues of the scattering matrix can be derived as $\lambda_{1,2} = t \pm \sqrt{r_L r_R}$.

Results. –

The broadband laser and CPA modes of A-PT symmetric structures. In this section, we propose an A-PT symmetric structure pictured in fig. 1(a) by using balanced

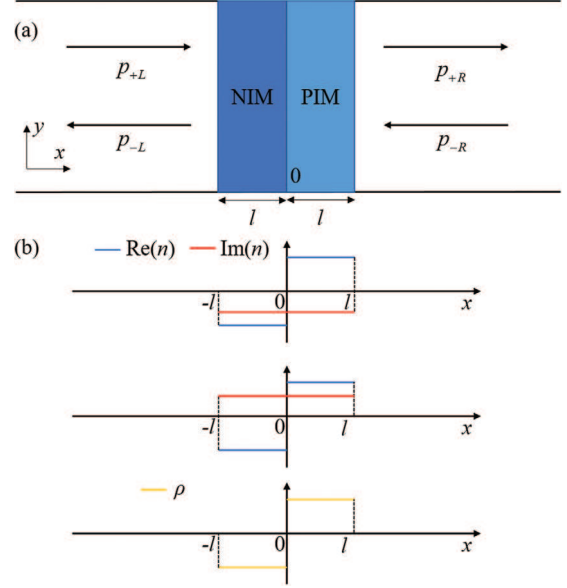


Fig. 1: The schematic picture, the refractive and density distributions of an A-PT symmetric structure. (a) Schematic picture of an A-PT symmetric structure by sticking a pair of balanced positive-index material (PIM) and negative-index material (NIM) together. The thickness of each layer is 50 mm. $p_{\pm L(R)}$ denotes the forward (backward) propagating acoustic waves in the left (right) side of the structure, respectively. (b) The refractive index and density distributions of the structure. The blue (red) solid lines show the real (imaginary) part of the refractive indices n of the structure, and the top panel illustrates an A-PT symmetric structure with loss, while the middle one pictures a structure with gain. The yellow solid lines in the bottom panel denote the density ρ .

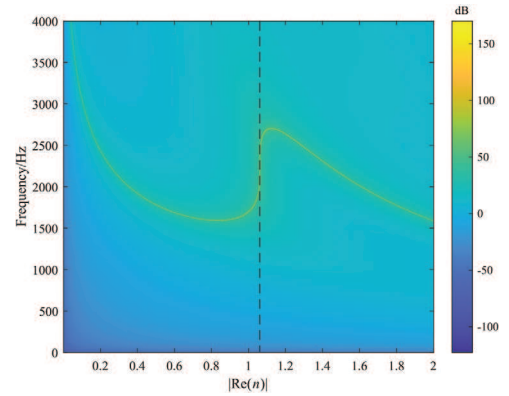


Fig. 2: The amplitude of the right-reflection coefficient of an A-PT symmetric structure with the variation of the real parts of the refractive indices and frequency. The imaginary parts are 0.4. The black dashed line denotes $|\text{Re}(n)| = 1.061$. The unit has been converted to decibels.

PIM and NIM, whose refractive index distribution satisfies $n(-x) = -n^*(x)$. Besides, the density distribution satisfies $\rho(-x) = -\rho(x)$. These conditions imply that the imaginary part of the refractive index is symmetric, which can be equal gain or loss, and the densities of PIM and

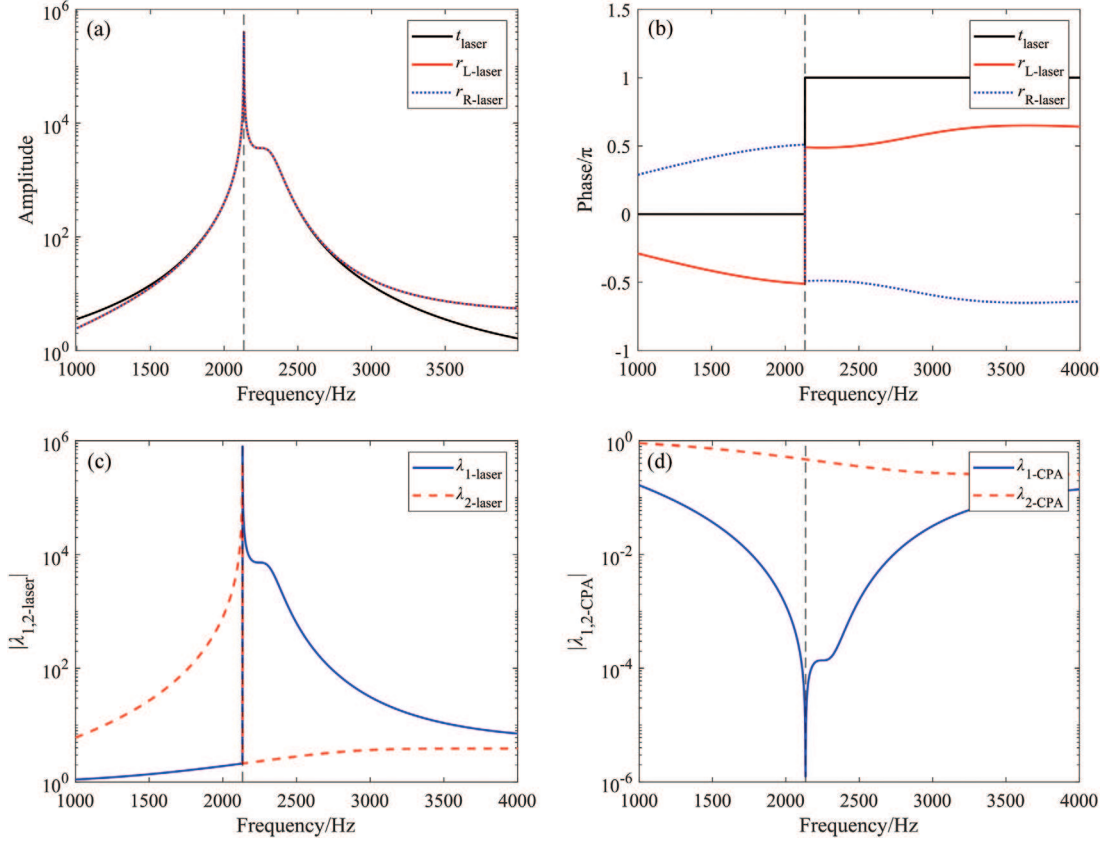


Fig. 3: Analytical results of the acoustic properties of the A-PT symmetric structure. (a) The amplitudes and (b) phases of the left-reflection, right-reflection coefficients and transmission coefficients of the A-PT symmetric structure with the refractive indices $\pm 1.061 + 0.4i$, which reaches its laser point at 2134Hz. (c), (d): the absolute value of the eigenvalues of the scattering matrix with the variation of frequency. Panel (c) is for the one with the refractive indices $\pm 1.061 + 0.4i$ and panel (d) is for the one with the refractive indices $\pm 1.061 - 0.4i$.

NIM are set as ρ_0 and $-\rho_0$, respectively, as is shown in fig. 1(b). For an A-PT symmetric structure, the elements of the transfer matrix must satisfy [34]

$$\text{Im}(T_{11}) = \text{Im}(T_{22}) = 0, T_{12} = -T_{21}^*. \quad (5)$$

Based on the transfer matrix, the coherent perfect absorption (CPA) mode occurs when $T_{11} = 0$ and the system achieves laser mode as $T_{22} = 0$. By flipping the sign of the imaginary part of the refractive index of the A-PT symmetric structure, which means replacing the acoustic gain of the structure with loss and vice versa, the transfer matrix $T = \begin{pmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{pmatrix}$ becomes $T' = \begin{pmatrix} T_{22} & -T_{12} \\ -T_{21} & T_{11} \end{pmatrix}$ (see details in the SM). It indicates that the laser mode of an A-PT symmetric structure will become a CPA one when the sign of its imaginary part of the refractive index is flipped while other parameters stay unchanged, and vice versa, because the CPA mode is a time-reversed counterpart of the laser mode. What is more, according to eq. (5), as T_{11} and T_{22} are both pure real, the solutions of $T_{11} = 0$ and $T_{22} = 0$ must be continuous in the parameter space, making it possible that the CPA and laser modes of an A-PT symmetric structure can be broadband. As is shown

in fig. 2, when the imaginary part of the refractive index is set as 0.4, and the real part of its refractive index is 1.061, the laser mode of the system is broadband.

In this case, the refractive indices of the A-PT symmetric structures are $\pm 1.061 + 0.4i$ for the one with a laser mode (corresponding to gain) and $\pm 1.061 - 0.4i$ for the one with a CPA mode (corresponding to loss). Figure 3(c) shows the absolute value of the eigenvalues of the scattering matrices for the structure with gain and fig. 3(d) is for the one with loss. It can be clearly seen that the two structures reach their laser point and CPA point at the frequency of 2134Hz, respectively, because the pole implies a laser mode while the zero here corresponds to a CPA mode [45]. At 2134Hz, the reflected and transmitted waves are amplified extremely and the phases of all the refraction and transmission coefficients of the one with gain experience an abrupt π jump, as depicted in figs. 3(a) and (b), leading to an acoustic laser. Furthermore, from figs. 3(c) and (d), the absolute value of one of the eigenvalues of the two structures is still extremely large (small) for the structure with gain (loss), indicating a broadbandness for the system investigated here.

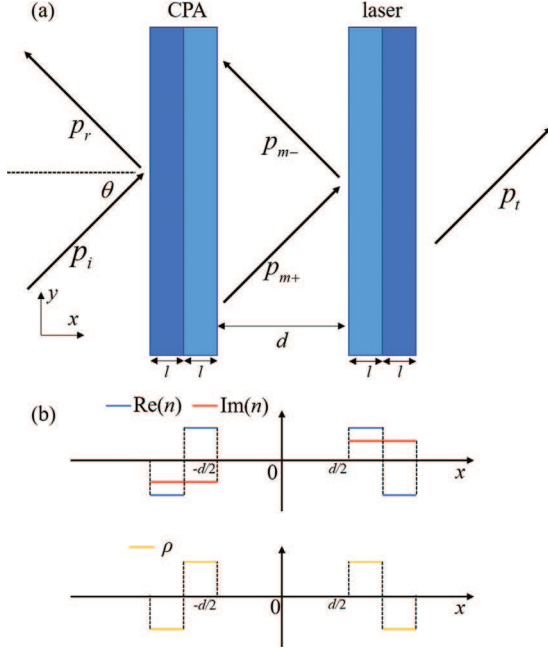


Fig. 4: The schematic picture, the refractive index and density distributions of the broadband negative refraction system. (a) Schematic of the broadband negative refraction system based on the laser and CPA modes of two A-PT symmetric structures. (b) The refractive index and density distributions of the system. The blue (red) solid lines show the real (imaginary) part of the refractive indices n of the structure and the yellow solid lines denote the density ρ .

Unidirectional broadband PT symmetric acoustic negative refraction based on the CPA and laser modes of A-PT symmetric structures. Inspired by the negative refraction systems based on PT symmetric structures [42,43], negative refraction can be achieved by the combination of the CPA and laser modes. As shown in fig. 4(a), the two metasurfaces are placed with a distance d between them, and the incidence angle is θ . The refractive indices of the left and right ones are $\pm 1.061 - 0.4i$ and $\pm 1.061 + 0.4i$, respectively. Specifically, transfer matrix of the total system with a normally incident wave from the left side at the laser (CPA) point 2134Hz can be derived as

$$T_{total} = \begin{pmatrix} e^{-ik_0 d} & -2iT_{11}T_{12} \sin k_0 d \\ 0 & e^{ik_0 d} \end{pmatrix}. \quad (6)$$

The transfer matrix shows that the left reflection coefficient is 0 and the transmission coefficient is $e^{-ik_0 d}$ according to eq. (3), indicating the unidirectional transparency of this system. And the acoustic pressure distribution in the middle of the system is

$$\begin{pmatrix} p_{m+} \\ p_{m-} \end{pmatrix} = \begin{pmatrix} 0 & -T_{12} \\ -T_{21} & T_{11} \end{pmatrix} \begin{pmatrix} p_i \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ -T_{21}p_i \end{pmatrix}, \quad (7)$$

p_{m+} and p_{m-} denoting the amplitudes of the forward propagating wave and backward propagating wave, respectively. Here, the T_{11} , T_{12} and T_{21} are all the elements of

the scattering matrix of the A-PT symmetric structure with gain. The result $p_{m+} = 0$ shows that the forward propagating wave between the two metasurfaces almost vanishes with a wave at the laser (CPA) point of the A-PT symmetric structure incident from the left side, implying the potential of negative refraction of the system.

As a matter of fact, the total system composed of two A-PT symmetric structures is PT symmetric because the density and refractive index distributions satisfy $\rho(-x) = \rho(x)$ and $n(-x) = n^*(x)$, as shown in fig. 4(b). Moreover, the exceptional point of the total PT symmetric system is also 2134 Hz, at which there is a transition from the PT symmetry-broken phase to the PT symmetric phase and there is a π jump in phase for its left-reflection coefficient, as depicted in figs. 5(a) and (b), respectively. Different from most of the PT symmetric systems learned before, the total PT symmetric system proposed here becomes unidirectionally reflectionless and transparent on a much wider bandwidth in frequency domain, ranging from 1850 Hz to 2600 Hz according to the behavior of its left-reflection and transmission coefficients depicted in figs. 5(b) and (c). The ratio of the amplitudes of the forward-traveling wave to the backward-traveling wave propagating between the two structures is pictured in fig. 5(d), which is smaller than 0.1 from 1500 Hz to 3200 Hz.

For oblique incidence, the parameters of the structure should be revised as $\rho(\theta) = \rho_{PIM(NIM)} / \cos \theta$ and $n_{PIM(NIM)}(\theta) = n_{PIM(NIM)} / \cos \theta_{PIM(NIM)}$ according to the incident angle θ . Here, $\theta_{PIM(NIM)}$ denotes the the complex refracted angle in the positive-(negative-) index material, which can be derived as $\theta_{PIM(NIM)} = \arctan(\sin \theta / n_{PIM(NIM)})$, and the detailed calculation can be seen in the SM. The numerical sound pressure distributions with different frequencies and incident angles with a harmonic plane wave incident from left on the system are shown in fig. 6, and it can be clearly seen that the reflectionless negative refraction works for different incident angles and different frequencies in this system. Some time-harmonic animations of the acoustic field distributions in fig. 6 are included as Supplementary Movies: [Animation_S1_2134_normal.gif](#), [Animation_S2_1850_oblique.gif](#), [Animation_S3_2134_oblique.gif](#), [Animation_S4_2600_oblique.gif](#).

In our further study, we are going to realize the system by using active sound-controlling apparatus with feedback and well-designed space-coiling structures [46,47] with loss. Considering the dispersion for the space-coiling structures, we could find a different refractive index range where the dispersion curves of the negative refraction system and the space-coiling structures are similar, instead of the dispersionless region we focus on here. For instance, figs. 7(a) and (b) show the acoustic properties of a negative refraction system consisting of NIM and PIM, of which the refractive index satisfies $|\text{Re}(n)| = 4.2144e^{-0.0006 \cdot f} + 0.3581$ and $|\text{Im}(n)| = 0.4$, and the results indicate that the reflectionless negative refraction

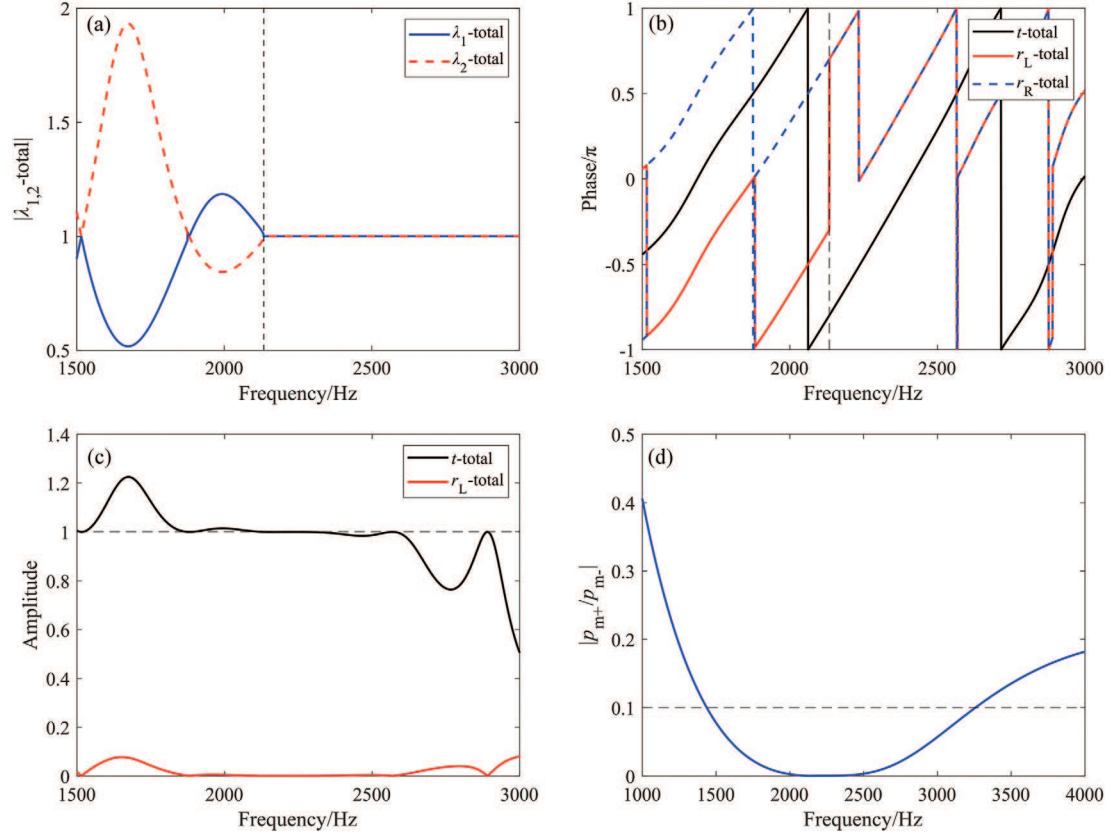


Fig. 5: Analytical results of the acoustic properties of the broadband negative refraction system. (a) The absolute value of the eigenvalues of the scattering matrix of the broadband PT symmetric negative refraction system combined with two A-PT symmetric structures. (b) The phases of transmission and reflection coefficients of the negative refraction system. (c) The amplitudes of the transmission and left-reflection coefficients of the negative refraction system. (d) The ratio of the amplitudes of the forward-traveling wave to the backward-traveling wave propagating in the middle area when there is a normal incidence wave from the left.

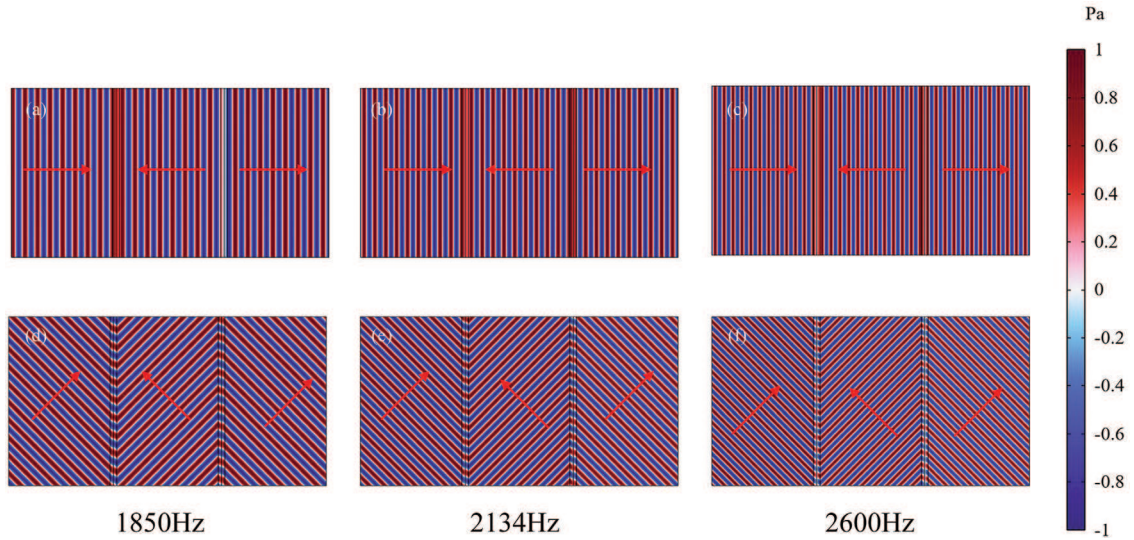


Fig. 6: Numerical results of the sound pressure distributions of a plane wave incident on the broadband negative refraction system at different angles and frequencies. The sound pressure distributions of a plane wave incident on the system at ((a), (d)) 1850 Hz, ((b), (e)) 2134 Hz, and ((c), (f)) 2600 Hz from the left side. The waves are incident normally in (a), (b) and (c) and incident at the angle of 45° in (d), (e) and (f). The amplitudes of the incident waves are all 1 Pa, and the red arrows represent the directions of the power flow.

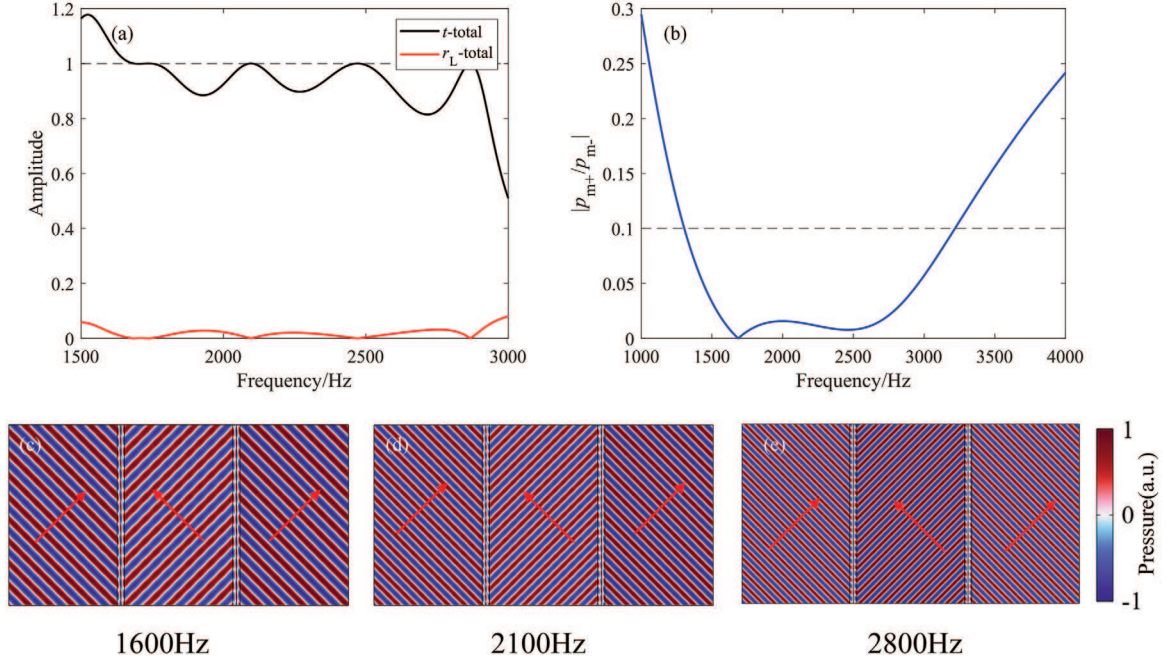


Fig. 7: Results of the acoustic properties of the negative refraction system consisting of dispersive material and the sound pressure distributions under left incidence. (a) The absolute value of the transmission and left-reflection coefficients of the negative refraction system. (b) The ratio of the amplitudes of the forward-traveling wave to the backward-traveling wave propagating in the middle area when there is a normal incidence wave from the left. In (c), (d) and (e), the plane waves are incident from the left at an incident angle of 45° , and the frequencies are 1600 Hz, 2100 Hz and 2800 Hz, respectively. The parameters of NIM and PIM have been revised according to the incident angle.

still works from 1600 Hz to 2800 Hz. In figs. 7(c), (d) and (e), the numerical results of the corresponding negative refraction at an incident angle of 45° are presented, in which the parameters have been revised according to the incident angle.

Discussion and conclusion. – In conclusion, we theoretically and numerically study the A-PT symmetric structure composed of a pair of balanced positive- and negative-index materials featuring equal gain/loss, which has a continuous laser or CPA spectrum. And with two parallel A-PT symmetric structures at their CPA and laser modes, a broadband reflectionless negative refraction system based on the continuous property of them is realized. After further research, this negative refraction system is found to be PT symmetric and its exceptional point corresponds to the CPA point and laser point of the two components, respectively. The proposed acoustic system here may offer a new method of designing different functional devices such as broadband acoustic lens and cloaks.

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Data availability statement: All data that support the findings of this study are included within the article (and any supplementary files).

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